

Two lever positions, one load

Compare effort and hand travel.

Explain what changes when the pivot moves nearer the load.

Recover Monday's lever rule

Point to the effort, pivot and load.

What might happen if the pivot moves near the load?

Choose to test, direct, observe or record.

Check the model before touching

Adult checks the lever, stable pivot and secured load.

Use a light load and a small lift target.

Keep hands clear of the pivot.

Never flick the lever or launch the load.

What stays the same?

The same light load

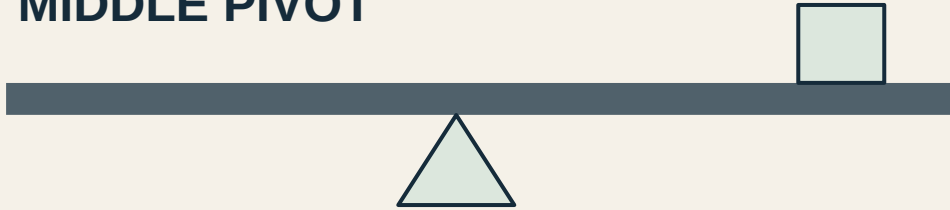
The same effort point

The same small height the load rises

The same way of judging or measuring effort

Compare the two arm lengths

MIDDLE PIVOT



PIVOT NEAR LOAD



effort at left • load at right

Nearer the load, the effort arm is longer relative to the load arm.

Predict: what will your hand do?

Try the middle pivot

Adult checks the middle pivot mark.

Raise the load to the small target height.

Notice the effort and the hand's travel.

Record what you observed.

Try the pivot nearer the load

Adult moves the pivot to its second mark.

Keep the same load, effort point and load rise.

Does it feel easier, similar or harder?

Compare hand travel with the first position.

Make your record precise

RECORD FOR BOTH POSITIONS

Same load/rise checked?
Effort: easier/similar/harder
Hand travel: what changed?

USE HONEST EVIDENCE

Observation is valid evidence.
Add cm only if measured.
Add N only with a force meter.

Pause: was the comparison fair?

Did the load stay the same?

Did it rise to the same target height?

Did the effort act at the same point?

Name one observation you want to check.

Repeat one careful comparison

Correct one setup or recording issue.

Compare both positions again if safe.

Keep any earlier observation and explain the change.

Explain the gain and the trade-off

With the pivot nearer the load, our model felt _____. The hand travelled _____.

For the same load rise, a longer effort arm can trade a smaller force for more travel.

Tidy and keep the evidence

Label EFFORT, PIVOT and LOAD on the sketch.

Keep observations beside your explanation.

Return the light load and all small parts.

What would help next time?

Did the equipment or waiting make learning harder?

Point, sign, speak, write—or pass privately.

Tell the adult teaching this lesson.

Agree one useful change and check whether it helps.

Show what you can explain

Supported: name the turning point of a lever.

Standard: why can a near-load pivot help lifting?

Stretch: explain why the benefit has a cost.

Check the force–distance trade-off

The pivot is the turning point. Moving it near the load can reduce the effort force.

The effort end travels farther. Real friction and bending can affect the result.